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A Data Driven Approach to Generating Synthetic Time-Lapse Casing Corrosion Logs

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Abstract

Downhole casing corrosion monitoring is crucial in production engineering to ensure well integrity and extend serviceability. Engineers perform corrosion time-lapse logs at intervals based on corrosion severity. With increasing well numbers and complexity, the demand for logs increases, challenging cost-effective surveillance. This paper introduces a machine learning application using One-Versus-Rest (OvR) Multi-class Classification to predict the next casing corrosion time-lapse log for previously logged wells.

This study investigates Galvanic corrosion in oil and water wells, which propagates originally from the outermost casing to inner casings. Corrosion severity is determined by the degree of metal loss and is typically categorized into high, medium, and low classes. A machine learning model is applied to classify inputs at each depth zone into one of these classes, generating a comprehensive corrosion log. The model architecture consists of binary classifiers based on Light Gradient-Boosting (LightGBM) algorithm, predicting class membership probabilities. The final corrosion class prediction is assigned based on the maximum probability among the classifiers.

Corrosion progression was found to be predominantly influenced by environmental exposure, particularly at interfaces between casing and surrounding formations. Factors such as corrosion history, structural attributes, depth-specific characteristics, and broader geological and geographical features were identified as primary contributors to model's performance. A machine learning framework was developed to estimate corrosion severity classification across depth zones and casings. The model demonstrated high accuracy in predicting metal loss growth over time in critical high corrosive depth zones, showcasing its potential to identify early signs of corrosion-related risks and avoid unnecessary logging costs, thereby providing a valuable tool for the oil and gas industry to optimize well maintenance and integrity.

The method presented a novel approach to optimizing and potentially reducing costly corrosion time-lapse logging requirements while ensuring the safety of operations. By leveraging AI to generate a portion of the time-lapse logs, production engineers can balance budgetary constraints while focusing physical logging efforts where they are most needed.

Key words: corrosion logs, well integrity, corrosion monitoring, time-lapse logs, well surveillance, artificial intelligence, machine learning, corrosion prediction

Introduction

The oil and gas industry rely heavily on the integrity of its wells to ensure safe and efficient production. However, as wells age and operate under increasingly complex environmental conditions, the risk of corrosion-related failures, leaks, or shut-ins becomes a significant concern. Casing corrosion, in particular, poses a major threat to well integrity, production sustainability, and asset life (Yakoot et al., 2021). The industry has traditionally relied on corrosion logging to monitor the condition of well casings, but this approach has its limitations, particularly high costs, and the sheer scale of existing well inventories.

The importance of Corrosion Monitoring

Corrosion monitoring is a critical component of well integrity management, as it enables operators to detect potential issues before they become major problems. Corrosion logs provide valuable insights into the condition of well casings, allowing operators to identify areas of high corrosion risk and prioritize maintenance and repair activities. However, these maintenance and repair activities are costly. According to Sircar et al. (2021), the application of machine learning and artificial intelligence in the oil and gas industry has shown significant potential in improving operational efficiency and safety, including in the area of corrosion monitoring.

The Need for Predictive Approaches. Given the high cost of traditional corrosion logging, there is a growing need for predictive approaches that can simulate corrosion logs with reasonable accuracy. By leveraging machine learning algorithms and historical data, operators can generate synthetic corrosion logs that provide valuable insights into corrosion risk and enable more effective resource allocation (Al-Qahtani et al., 2025). This approach has the potential to revolutionize corrosion management in the oil and gas industry, enabling operators to prioritize logging interventions, optimize surveillance strategies, and reduce unproductive logging jobs. According to Macêdo et al. (2024), machine and deep learning techniques have been successfully applied to predict deformations in oil and gas pipelines, highlighting the potential of these approaches in corrosion management.

The Role of Machine Learning in Corrosion Risk Management. Machine learning has emerged as a powerful tool in corrosion risk management, enabling operators to analyze complex data sets and identify patterns and trends that inform corrosion prediction models (Fang et al., 2023). By integrating machine learning with traditional corrosion modeling techniques, operators can enhance the accuracy and reliability of corrosion prediction models, enabling more effective decision-making and resource allocation. According to Seghier et al. (2022), ensemble learning techniques have been successfully applied to predict internal corrosion rates for oil and gas pipelines, achieving high accuracy and robust results. This highlights the potential of machine learning approaches in reducing the cost and improving corrosion prediction and management in the oil and gas industry.

The Potential Benefits of Machine Learning

The application of machine learning in corrosion risk management has the potential to yield significant benefits, including improved safety, reduced costs, and enhanced decision-making (Madamanchi et al., 2024). By providing operators with accurate and reliable predictions of corrosion risk, machine learning models can enable more effective resource allocation, prioritize logging interventions, and optimize surveillance strategies. Additionally, machine learning can enhance the integration of geospatial analytics in corrosion risk management, enabling operators to identify clusters of wells with elevated corrosion risk and prioritize logging interventions accordingly. According to Yakoot et al. (2021), well integrity management in mature fields requires a state-of-the-art approach to address the unique challenges posed by aging infrastructure, and machine learning has the potential to play a critical role in this effort.

The Importance of Data Quality. The success of machine learning approaches in corrosion risk management depends on the quality of the data used to train and validate the models (Sircar et al., 2021). High-quality data is essential for developing accurate and reliable corrosion prediction models, and operators must ensure that their data management systems are capable of collecting, storing, and analyzing large amounts of data. According to, a risk-based inspection methodology is essential for ensuring the integrity of oil and gas facilities, and machine learning can play a critical role in this effort by providing operators with accurate and reliable predictions of corrosion risk.

The Future of Corrosion Risk Management. The future of corrosion risk management in the oil and gas industry is likely to be shaped by the increasing use of machine learning and artificial intelligence (Fang et al., 2023). As the industry continues to evolve and adapt to new challenges and opportunities, the application of machine learning is likely to play a critical role in improving corrosion prediction and management. According to Macêdo et al. (2024), machine and deep learning techniques have the potential to revolutionize corrosion management in the oil and gas industry, enabling operators to prioritize logging interventions, optimize surveillance strategies, and reduce unproductive logging jobs. This paper presents a machine learning-based approach to generating synthetic time-lapse casing corrosion logs, with the objective of providing production engineers with a forecasting tool that can predict future corrosion severity across casing depths.

Problem Statement

The problem of casing corrosion can be summarized as follows:

- Casing corrosion is a major threat to well integrity, production sustainability, and asset life (Yakoot et al., 2021)
- Current methods of corrosion monitoring are limited by their high costs
- There is a need for a more efficient and cost-effective method of corrosion monitoring and prediction

Solution Overview

This paper proposes a solution to the problem of casing corrosion by using machine learning algorithms to generate synthetic time-lapse casing corrosion logs. The solution involves:

- Using machine learning algorithms to analyze historical data and predict future corrosion severity
- Generating synthetic corrosion logs that provide valuable insights into corrosion risk and enable more effective resource allocation
- Providing production engineers with a forecasting tool that can predict future corrosion severity across casing depths

Methodology

This study employed a data-driven approach to predict the next corrosion time-lapse log for a previously logged well using machine learning techniques. It also investigates Galvanic corrosion in oil and water wells. The experimental procedure involved collecting and preprocessing historical corrosion logs data, engineering relevant data features, and developing a supervised machine learning framework. Additionally, domain-specific features related to Galvanic corrosion were incorporated into the analysis to assess their impact on the model's accuracy. The sub-sections outlined below describe the methodology followed throughout the study.

Data Collection and Description

The dataset utilized in this study consists of historical corrosion logs covering oil and water wells. It included depth-based features that might have a direct or indirect relation to corrosion such as pressure, temperature, flow parameters, and well-specific parameters. The data provided sufficient variability to support the development of the methodology to predict the next corrosion time-lapse log.

Data Preprocessing

Prior to model training, several preprocessing steps have been applied to the data to ensure quality and consistency. Missing values were imputed using data points from offset wells located within 5KM radius, allowing the data to preserve contextual patterns across wells with similar characteristics. Categorical features were transformed through one-hot encoding to facilitate their integration into the machine learning framework.

Machine Learning Framework

To enable generation of a comprehensive corrosion log per depth and casing size, we framed the task as a multi-classification problem targeting metal loss severity. The target variable was derived from historical metal loss percentages acquired from corrosion logging tools over the years, which we discretized into three corrosion classes: high, medium, and low – each representing a distinct range of metal severity.

Our end-to-end framework, illustrated in Figure 1, included data collection, preprocessing, class imbalance handling, dimensionality reduction, model training, and log generation.

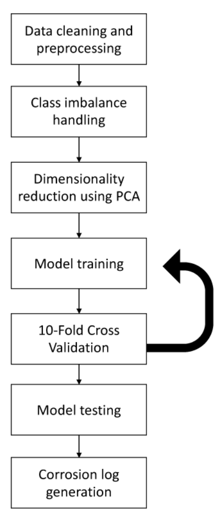

Figure 1—A general ML framework to generate a synthetic corrosion log

Following an extensive evaluation of various machine learning algorithms including neural networks, tree-based models, and ensemble techniques we adopted a One-versus-Rest (OvR) classification strategy as the final modeling approach as illustrated in Figure 2. This allowed us to train a dedicated binary classifier for each severity class, improving the model's ability to distinguish between overlapping patterns in corrosion behavior.

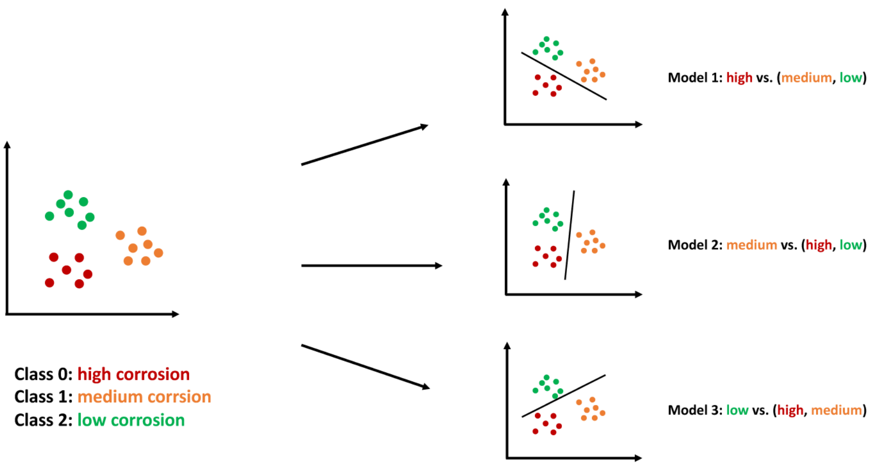

Figure 2—One-vs- Rest Framework used to decompose a multi-class problem into binary classifiers.

Evaluation Metrics and Validation

We evaluated the models' performance using 10-fold cross validation technique. In this approach, the dataset is split into 10 folds where the model is trained on 9 folds and tested on the remaining one. Repeating the process ten times so that each fold serves as the test set once, the results are then averaged to provide a more stable estimate of model performance.

All models were evaluated using standard classification metrics: accuracy, precision, recall, and F1-score. These provided a well-rounded view of performance, particularly important in a multi-class setup with class imbalance. In addition, we used the confusion matrix to gain further insight into class-level misclassifications.

Prediction Process

Once the OvR framework is trained, the prediction for a new data point follows a systematic process. Since the problem is formulated as a three-class classification task (high, medium, and low corrosion severity), three independent binary LightGBM classifiers are trained, each responsible for distinguishing once class against the rest.

When a new data point is introduced during prediction, it is passed through all three classifiers simultaneously:

Probability Estimation. Each binary classifier outputs a probability scope that the instance belongs to its designated class. For example:

- $P_{high} = f_{high}(x)$
- $P_{medium} = f_{medium}(x)$
- $P_{low} = f_{low}(x)$

Where $f_{class}(x)$ represents the probability predicted by the classifier trained for that class.

Decision Rule. To determine the final class assignment, the three probabilities are compared, and the class with the maximum probability is selected. This ensures that the prediction corresponds to the classifier most confident in its assessment.

$$\hat{y} = \quad argmax_{c \in \{high, medium, low\}} P_c$$

Final Output. The selected class label (high, medium, or low) is then reported as the predicted metal loss severity for the new data point. By structuring predictions in this way, the OvR approach ensures consistent decision-making across all classes, while still benefiting from the strength of specialized binary classifiers.

Results

Cross-Validation Results

To accurately evaluate the performance of the model, we applied a 10-fold cross validation technique across all candidate algorithms. We evaluated the performance of each binary classifier individually as part of the OvR technique used. The main evaluation metrics included accuracy, precision, recall, and F1-score, all averaged across all folds.

Table 1 summarizes the performance (Mean Recall Score) of all tested models, including our final selected model, the LightGBM OvR classifier, evaluated across the three corrosion severity classes.

Table 1—Cross Validation accuracy per corrosion severity classifier for all models (OvR technique)

Model	High vs (Medium, Low)	Medium vs (High, Low)	Low vs (High, Medium)
Decision Tree	82%	74%	94%
Random Forest	82%	83%	95%
XGBoost	83%	83%	90%
ANN	84%	79%	73%
LighGBM	84%	84%	91%
Gradient Boost	83%	82%	91%
KNN	55%	21%	67%
Logistic Regression	75%	20%	35%
AdaBoost	49%	28%	24%
Bagging	49%	30%	65%

The LightGBM OvR model achieved the highest recall score across all three classes, with particularly strong performance in detecting high corrosion zones. While most models struggled to separate medium from the adjacent classes—likely due to overlapping metal loss features. Non-tree-based algorithms such as Artificial Neural Network (ANN) demonstrated competitive predictive accuracy but tended to be overly biased toward predicting the high corrosion class, resulting in reduced recall for the medium and low classes. LightGBM consistently delivered balanced results and lower misclassification rates, as later confirmed by confusion matrix results on the held-out dataset.

These findings validated our selection of LightGBM as the final model for full-scale testing and corrosion log generation.

Final Model Testing on Held-Out Data

To validate the generalization capability of the selected model, we performed blind testing on a held-out dataset composed of recent corrosion logs from the year 2024 and 2025. This dataset was excluded entirely from the training and cross-validation process and served as an independent benchmark to confirm the model's real-world applicability.

One of the main aspects of testing was the model's ability to detect corrosion progression between the previous log and the one we are generating, particularly where the corrosion class shifts from medium or low corrosion to high corrosion. Identifying such scenario is a significant operational interest for early intervention and risk mitigation in well integrity management.

The final selected LightGBM OvR model demonstrated balanced predictive performance on unseen data. Additionally, it successfully detected the phenomenon of interest where corrosion severity progressed over the years.

Figure 3 shows the confusion matrix for the model's predictions on the held-out dataset, illustrating class-wise accuracy. The model achieved a recall of 90% for high corrosion, 69.23% for medium corrosion, and 86.48% for low corrosion. These results are closely aligned with cross-validation results, confirming the model's consistency and robustness.

True Label \ Predicted Label	High corrosion	Medium Corrosion	Low Corrosion
High Corrosion	90%	5%	5%
Medium Corrosion	19.23%	69.23%	11.54%
Low Corrosion	5.41%	8.11%	86.48%

Figure 3—Confusion Matrix of final model results on held-out dataset from 2024 and 2025.

Conclusion

This paper introduced a machine learning framework for generating synthetic corrosion time-lapse logs, aimed at optimizing corrosion re-logging requirements. By translating the business challenge into a multi-class classification problem to classify metal loss severity into three categories – high, medium, and low. We addressed the problem using an OvR classification approach, with LightGBM selected as the final model based on consistent high performance across cross-validation and blind testing.

The results demonstrated the model's ability to not only generalize on unseen data but also to capture corrosion progression over time. This capability highlights the potential of AI-driven solutions to support predictive maintenance, optimize physical logging campaigns, and enable proactive well integrity management. From an operational standpoint, this framework directly contributes to cost savings and improved resource allocation, all of which are critical priorities to navigate increasingly complex field conditions. On the technical side, the study validated the robustness of OvR approach in handling extremely imbalanced datasets and achieving interpretable results across multiple metal loss severity classes.

Looking ahead, further enhancements will focus on scaling the solution to additional oil fields, and expanding the study to cover gas fields where corrosion dynamics differ slightly. Future research could also explore incorporating additional features related to cement quality, which has a direct influence on corrosion risk and could further improve the predictive capability of the framework. Furthermore, hybrid modeling approaches could be explored such as combining supervised learning with physics-informed models to further improve reliability.

In summary, this framework provides a promising step toward scalable, data-driven well surveillance. It presents a novel approach to optimizing and potentially reducing costly corrosion time-lapse logging requirements while ensuring the safety and integrity of operations. By leveraging AI to generate a portion of the time-lapse logs, production engineers can balance budgetary constraints while focusing physical logging efforts where they are most needed.

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